

Exercise

Given the independent and dependent variables, manipulate the original equation to the form $y = mx + c$, and state the quantities to be plotted and the expression for the gradient and the y-intercept.

The quantities y and x should only consist of the independent and dependent variables.

Independent & Dependent Variables	Original Equation	To the Form $y = mx + c$	Quantity y	Quantity x	Gradient m	y-Intercept c
d, T	$\frac{2\pi}{T} = \sqrt{\frac{3g}{h}} \times \frac{d}{L}$	$\frac{1}{T} = \frac{1}{2\pi L} \sqrt{\frac{3g}{h}} \times d$	$\frac{1}{T}$	d	$\frac{1}{2\pi L} \sqrt{\frac{3g}{h}}$	0
h, T	$T^2 = \frac{kML^2}{12mh} + kh$	$T^2h = kh^2 + \frac{kML^2}{12m}$	T^2h	h^2	k	$\frac{kML^2}{12m}$
M, T	$T = 2\pi \sqrt{\frac{M+m}{k}}$	$T^2 = \frac{4\pi^2}{k}M + \frac{4\pi^2m}{k}$	T^2	M	$\frac{4\pi^2}{k}$	$\frac{4\pi^2m}{k}$
L, T	$T = kL^n$	$\lg T = n \lg L + \lg k$	$\lg T$	$\lg L$	n	$\lg k$
R, V	$E = V + \frac{ER}{X+R}$	$\frac{1}{V} = \frac{1}{EX} \times R + \frac{1}{E}$	$\frac{1}{V}$	R	$\frac{1}{EX}$	$\frac{1}{E}$
L, R	$R = kL \left(1 - \frac{L}{L_0} \right)$	$\frac{R}{L} = -\frac{k}{L_0} \times L + k$	$\frac{R}{L}$	L	$-\frac{k}{L_0}$	k
t, R	$R = R_0 e^{-\frac{t}{\tau}}$	$\ln R = -\frac{1}{\tau} \times t + \ln R_0$	$\ln R$	t	$-\frac{1}{\tau}$	$\ln R_0$
θ, I	$\frac{\mu_0 I}{2\pi h} = B_0 \sin \theta$	$I = \frac{2\pi h B_0}{\mu_0} \times \sin \theta$	I	$\sin \theta$	$\frac{2\pi h B_0}{\mu_0}$	0